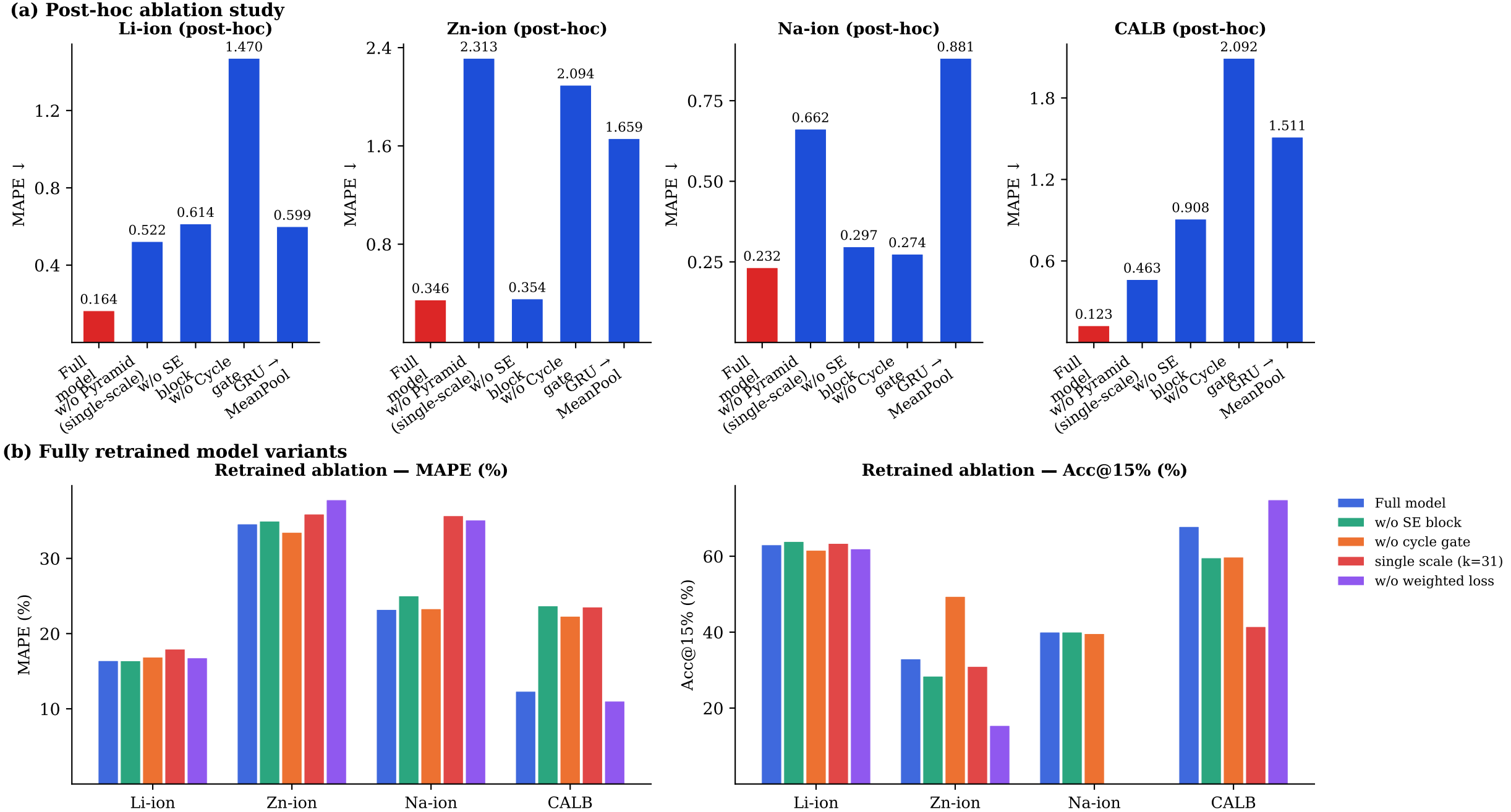
(a) Post-hoc ablation study
Li-ion (post-hoc)
Zn-ion (post-hoc)
Na-ion (post-hoc)
CALB (post-hoc)
MAPE ↓
0.164
0.522
0.614
1.470
0.599
0.346
2.313
0.354
2.094
1.659
0.232
0.662
0.297
0.274
0.881
0.123
0.463
0.908
2.092
1.511
Full model
w/o Pyramid (single-scale)
w/o SE block
w/o Cycle gate
GRU → MeanPool
(b) Fully retrained model variants
Retrained ablation — MAPE (%)
Retrained ablation — Acc@15% (%)
MAPE (%)
Acc@15% (%)
Li-ion
Zn-ion
Na-ion
CALB
Full model
w/o SE block
w/o cycle gate
single scale (k=31)
w/o weighted loss